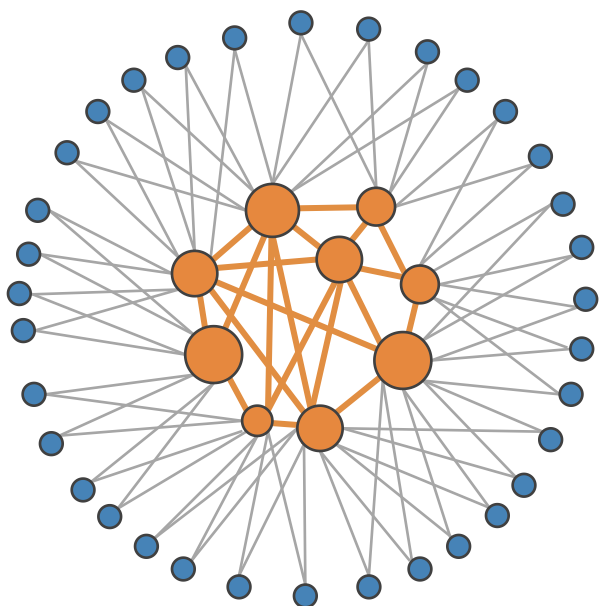


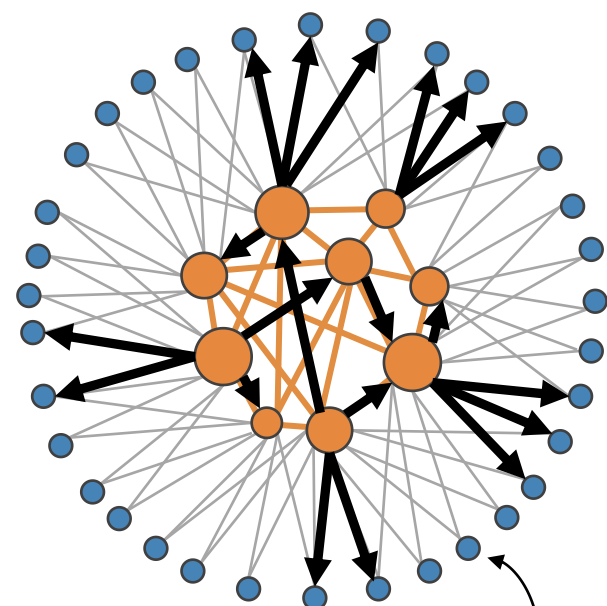
A True population (what we want)

Community A Community B

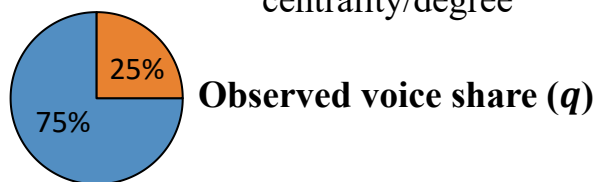


Designer goal: each person counts equally.

B How feedback gets collected (what happens in practice)



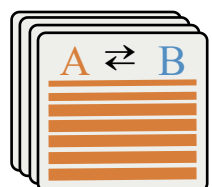
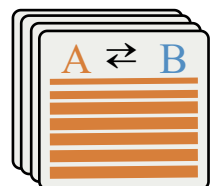
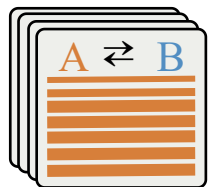
$q(i)$ increases with centrality/degree



Network sampling → high-degree nodes appear more often.

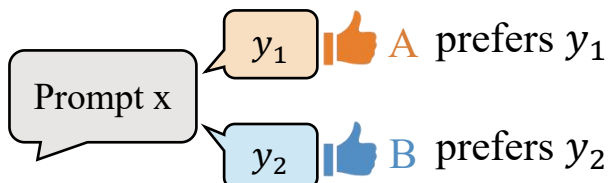
C What standard RLHF does with the data

Preference Dataset



Single reward model

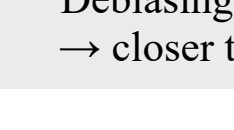
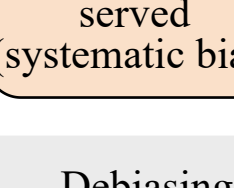
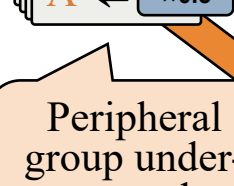
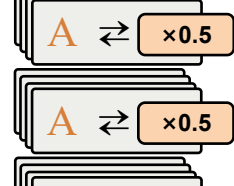
Policy update



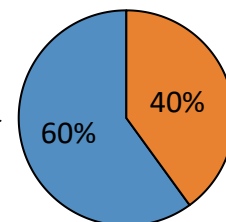
Standard RLHF averages what it sees → effectively optimizes q -weighted preferences.

D Graph-Preference Learning

Preference Dataset



Reweighted voice share ($\tilde{\pi}$)



Peripheral group under-served (systematic bias)

Policy selects y_2 or compromise y^*

Debiasing + graph sharing → closer to target welfare.

Weights: $\alpha_i \approx \pi_i / q_i$